

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Evaluating generative models demands human judgment, because standard metrics fail to reflect human preferences. Human evaluations are far more reliable, but they are costly and noisy: participants vary in expertise, attention, and diligence. Pairwise comparisons sharpen consistency, yet turning them into overall quality scores requires principled modeling. Bradley-Terry-based methods update item scores from comparisons, but existing approaches fall short on both fronts—they either ignore rater variability or lack convergence guarantees—which undermines their robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry variant that explicitly models rater quality, downweighting or removing unreliable participants, and delivers guaranteed monotonic likelihood convergence through an Expectation-Maximization algorithm. Across noisy and crowdsourced settings, BBQ converges faster, produces well-calibrated uncertainty estimates, and yields substantially more robust and interpretable rankings than baseline Bradley-Terry models. This framework makes human evaluation of generative models both more reliable and more cost-effective.

1 INTRODUCTION

Evaluating generative models is a hard problem, particularly for large language models (LLMs) and image generators, where standard metrics fail to reflect human preferences. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, capture only a narrow slice of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluation is therefore indispensable for establishing meaningful rankings between models. This is precisely why leaderboards and comparative studies of large language models and Learned Image Compression (LIC) methods rely so heavily on human preference data.

Human evaluation, however, is expensive and time-consuming, so the protocol must be efficient while minimizing subjectivity and noise. Pairwise comparisons—asking participants to choose between two items rather than assign absolute scores—are the most effective and practical form of human evaluation for this purpose (Zerman et al., 2018; Wang et al., 2023). They are easier for participants and yield markedly more consistent judgments. Collecting all pairwise comparisons is infeasible, because the number of pairs grows quadratically with the number of items. And with only a limited set of comparisons, simple statistics such as win rates cannot recover overall quality scores, because a win rate depends on the quality of the items it is compared against.

Producing an overall ranking for a leaderboard therefore demands an effective aggregation of pairwise comparisons, which in turn calls for aggregation methods that are both robust and efficient. The Bradley-Terry (BT) (Bradley & Terry, 1952) model meets this need by iteratively updating item scores from comparison outcomes, and it has become the standard approach for aggregating pairwise judgments across domains.

For all its usefulness, the standard Bradley-Terry model carries two decisive limitations in practice. BT parameters can be estimated with several algorithms, including iterative scaling methods (Ford Jr, 1957), gradient descent, and the minorization maximization (MM) algorithm (Hunter, 2004). The MM algorithm guarantees convergence to the global maximum for the basic BT model; gradient-based methods, which are closely related to the Elo rating system, are widely used yet offer no such guarantee (Hunter, 2004). Worse, for extended BT models that incorporate factors such as

home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). The second limitation is that human evaluations are inherently noisy: participants rush, guess, or lose focus, so reliability varies sharply across raters. Ignoring this variability lets unreliable raters distort item scores and destabilize the ranking.

We resolve both limitations with a Bayesian Bradley–Terry variant that jointly models item quality and rater reliability. To the best of our knowledge, this is the first method to combine the BT framework with a Bayesian formulation that explicitly models rater quality while providing closed-form EM updates. Previous approaches to noisy raters rely on gradient-based optimization without convergence guarantees; our method instead guarantees stable convergence and yields interpretable parameters. The Bayesian framework directly addresses epistemic uncertainty and supplies principled regularization. To capture variability in participant behavior, we introduce rater-specific parameters that quantify how consistent and trustworthy each participant is, letting the model adjust the influence of individual comparisons accordingly. We derive an EM algorithm that efficiently estimates item and rater parameters through a latent-variable formulation and guarantees monotonic likelihood improvement. This lets partially reliable raters still contribute meaningful information while sharply reducing the impact of inattentive or inconsistent participants. By modeling rater quality within a Bayesian framework, our method further delivers uncertainty estimates for item scores, enabling more interpretable rankings and robust comparisons across studies.

We establish the effectiveness of our approach on multiple human evaluation datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. Our method converges faster, is markedly more robust to noisy comparisons, and produces more consistent rankings than standard BT and naive aggregation methods. The framework reaches well beyond generative model evaluation: it applies to any setting where noisy pairwise comparisons must be turned into reliable global rankings. In short, our work delivers more cost-efficient, interpretable, and reproducible human studies for evaluating AI-generated content.

2 RELATED WORK

Human studies routinely require ranking items. Presenting participants with a choice between two items rather than a single item with a score sharpens sensitivity to subtle differences and cuts variability in responses (Zerman et al., 2018; Wang et al., 2023). For this reason, most leaderboards for machine learning models rely on pairwise comparisons: participants indicate their preference between two alternatives, and these preferences are aggregated into overall rankings. The Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025) are leading examples, both collecting pairwise comparisons and combining them with variants of the Bradley–Terry (BT) model.

The Bradley–Terry model (Bradley & Terry, 1952), originally developed to rank competitors in sports, provides a probabilistic framework for estimating the likelihood that one item is preferred over another. By converting pairwise comparisons into a ranking, it has become a cornerstone of studies across games, consumer preferences, and many other applications. A range of extensions broaden its reach. The Plackett–Luce model generalizes BT from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations by multiplying successive BT probabilities (Luce et al., 1959). Other variants target specific contexts, modeling home-field advantages (Agresti, 2010), incorporating ties (Rao & Kupper, 1967), or handling comparisons between groups rather than individuals (Huang et al., 2006).

The parameters of the basic BT model are typically inferred by maximum likelihood estimation (MLE). Zermelo (1929) introduced the iterative approach that computes these estimates and remains widely adopted today. Lange et al. (2000) later showed that this procedure is a special case of minorization-maximization (MM) algorithms, which iteratively optimize surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM algorithms to generalized BT models and established the conditions that guarantee convergence to the MLE. For the classical BT model the optimization is convex, so convergence to the global optimum is assured (Hunter,

¹The IHQ dataset will be released upon acceptance.

2004). For extensions such as home-field advantage, multiple comparisons, or ties, however, MM algorithms may converge only to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model incorporate prior knowledge and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference via Gibbs sampling. This formulation yields tractable complete-data likelihoods and guarantees convergence of the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior estimates, reduces susceptibility to local maxima, and quantifies uncertainty in the estimated item skills.

Annotator quality models are the critical extension that abandons the untenable assumption that all comparisons are equally reliable. Because participant quality varies widely in crowdsourced evaluations, several approaches explicitly account for annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters. Their approach places Gaussian priors on item and rater parameters and inserts scaling factors in the likelihood to modulate each annotator’s influence. Crucially, they perform posterior inference with gradient-based optimization, which struggles in high-dimensional spaces because of local optima and, more fundamentally, carries no convergence guarantee. Our EM-based method, by contrast, guarantees monotonic likelihood improvement. The Elo-based rater model developed by Google Research implements a simplified version of this idea and is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, ours is the first method to combine the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates. Where Chen et al. (2013) rely on Gaussian priors and gradient-based optimization without convergence guarantees, we exploit conjugate priors and an EM algorithm that guarantees monotonic likelihood improvement. And where the Elo-based rater model used in CLIC (CLIC, 2025) is a simplified heuristic built on Elo-style updates, our method offers a principled Bayesian treatment with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Converting a set of noisy pairwise comparisons into a reliable ranking of items is a fundamental problem in machine learning and statistics. Consider a set of K items that are repeatedly compared in pairs by a set of R raters. The data, which we denote as D , consists of the outcomes of these comparisons. For two items i and j of this set, Bradley & Terry (1952) suggested the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating. This model gives a clear and interpretable way to infer item scores from observed wins and losses. It rests, however, on the simplifying assumption that all comparisons are equally reliable—an assumption that human evaluation routinely violates. Participants differ in expertise, attention, and diligence, and these differences produce unreliable and inconsistent judgments.

To capture varying rater reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. With probability q_r , rater r makes an informed judgment following the Bradley-Terry model; with probability $1 - q_r$, the rater guesses randomly, as if flipping a fair coin between the two items. This yields the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\log P(D \mid \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . This likelihood admits no closed-form maximum likelihood estimator, so the optimal parameters λ and q cannot be derived analytically. An iterative optimizer such as gradient descent could be applied, but it offers no guarantee of convergence. We instead adopt the Expectation–Maximization (EM) algorithm, which eliminates learning-rate tuning and guarantees a monotonic increase of the observed-data likelihood together with convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we interpret the Bradley-Terry model under a Thurstonian framework (Diaconis, 1988). Here a comparison between items i and j is a race, in which each item has a random arrival time, Y_i and Y_j . These arrival times follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time wins. This yields the standard Bradley-Terry probability directly:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables to simplify the complete-data likelihood. First, we define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which denotes whether the c -th comparison of items i and j by rater r follows the Bradley-Terry model. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent variable is the key that makes the EM update tractable while still accounting for rater-specific quality.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm finds maximum a posteriori (MAP) estimates of our model parameters λ and q by treating rater quality and the unobserved Thurstonian arrival times as latent variables, and it is guaranteed to converge to a stationary point of the posterior distribution.

We first specify prior distributions for the parameters. The item skills λ_k receive a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, the conjugate prior for the exponential distribution, and each rater’s quality parameter q_r receives a Beta prior, $q_r \sim B(\alpha, \beta)$.

The algorithm iteratively maximizes the expected complete-data log-posterior conditioned on the current parameter estimates (λ^*, q^*) , alternating between two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, denoted Q . This function is the expected value of the log-posterior over all observed and latent variables, given the observed data and the parameter estimates from the previous iteration. It is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , is further broken down into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization This step updates the model parameters by maximizing the Q function. Leveraging the expected values from the E-step, the M-step reduces the original hard optimization problem to simple, closed-form updates. Both updates hinge on a single quantity: the posterior probability that a given comparison from rater r follows the Bradley-Terry model. This quantity, denoted γ , is the weight of a rater’s judgment, set by how closely it aligns with the model’s current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . The posterior probability γ directly encodes our confidence that a comparison was meaningful rather than random, given the current parameter estimates: higher γ means a more trustworthy comparison.

The new estimate of a rater’s quality q_r is a weighted average. The numerator sums the “effective number of wins” for that rater, each win weighted by the posterior probability γ that it was a meaningful, non-random judgment, and combines this with the Beta-prior hyperparameters to regularize the estimate. The denominator normalizes by the total number of comparisons and the prior parameters. The update thus raises a rater’s quality score precisely when their judgments align with the model’s predictions:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The new estimate of an item’s skill λ_i is a ratio balancing two quantities. The numerator sums the “effective wins” for item i across all raters, again weighting each win by the rater’s quality γ ; this term captures the total positive evidence for item i . The denominator accounts for all comparisons in which item i participated and acts as a normalizing factor. Both terms are regularized by the Gamma-prior hyperparameters. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The full derivation of the Expectation-Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate our **Bayesian Bradley-Terry model with rater Quality (BBQ)**, we benchmark it against two strong baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient descent-based BT model that incorporates rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025), the same one employed by both CLIC (2025) and Ballé et al. (2025). The hyperparameters used in our experiments are listed in Section C.

We evaluate several Bradley-Terry (BT) variants across a broad range of datasets: human preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023))

and for image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater quality, we partition the IHQ dataset into screened and unscreened subsets, which lets us measure directly how low-quality, high-quality, and mixed comparisons affect model performance. Details of the IHQ dataset creation are given in Section E, and a summary of all datasets—comparisons, raters, and items—is provided in Section D and Table 2.

Ground truth rankings are unavailable for real datasets. We therefore first validated the models on simulated datasets, where all methods recover the same ordering as the number of samples grows. For crowd-sourced datasets, we approximate the ground truth by the ranking obtained on the whole dataset, and we confirm that this is a sound approximation of the true ordering by examining the top-1 item in each ranking. For every dataset except IHQ-unscreened, all three models recover the same top-ranked item. On IHQ-unscreened, however, Crowd-BT and Bayes-BT both fail to identify the reference image as the highest-quality item, while BBQ succeeds.

A reliable aggregation method must reproduce the same ordering when the study is repeated. We quantify this stability with bootstrapping, which estimates the variability in the rankings; the procedure is described in Section F. Note that the WD and HiFiC studies used active selection of comparison pairs—HiFiC via a binary search strategy and WD via maximum information gain—so the bootstrapping results on these two datasets should be interpreted with caution.

We report two complementary metrics. Top-1 agreement is the fraction of bootstrapped samples that identify the same best item as the full dataset. Kendall’s Tau (τ) measures the ordinal correlation between rankings (Kendall, 1938), where higher values indicate more stable results. Top-1 agreement matters most when the single best model is the target, whereas Kendall’s Tau assesses overall ranking stability.

4.1 PERFORMANCE ACROSS DATASETS

We compare the models across datasets on ranking accuracy and stability. Top-1 agreement and Kendall’s Tau are summarized in Table 1. Datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are generally easier. HUMAINE breaks this trend, a clear reminder that factors beyond the raw number of comparisons—rater consistency and diversity among them—drive model performance.

BBQ delivers the best stability and robustness across datasets. It identifies the top-performing item most frequently, tying for best on three datasets. And it recovers the overall ranking most accurately on five of eight datasets, placing second on the remaining three.

Tellingly, the three datasets where BBQ ranks second in Kendall’s Tau are exactly those with high-quality, homogeneous rater sets. On MT-Bench, all models perfectly recover the top item. The WD dataset, collected by Ballé et al. (2025), includes only five raters, likely the paper’s authors, indicating exceptionally careful evaluation. In IHQ-screened, raters were explicitly filtered for quality. When rater quality is uniformly high, there is simply little variability left to model, so explicitly modeling rater reliability yields little advantage—precisely the regime in which BBQ’s edge over simpler models is expected to shrink.

The IHQ experiments, spanning both screened and unscreened rater subsets, reveal an unmistakable pattern. BBQ substantially outperforms Crowd-BT and Bayes-BT whenever low-quality raters are present, as in CLIC-all and CLIC-unscreened. All three models reach their best Top-1 accuracy on the screened subset, underscoring the value of rater selection. Crowd-BT models rater quality explicitly, yet its performance drops sharply on the full dataset—almost certainly because crowdsourced raters provide fewer than 40 comparisons each, leaving it exposed to noisy annotations. BBQ, by contrast, holds its performance even without screening, losing only marginal Top-1 accuracy and thereby demonstrating genuine robustness to low-quality raters.

Achieving uniformly high rater quality, however, is rarely feasible. Large-scale crowdsourcing inevitably introduces variability, screening procedures are costly, and subjective factors sway even diligent annotators. In exactly these conditions, BBQ offers a principled way to exploit partially reliable raters while suppressing the impact of noisy contributions.

Table 1: Performance of BT-based aggregation methods across datasets. Top: Top-1 agreement [%]; Bottom: Kendall’s Tau. BBQ identifies the top-performing item most frequently, recovers the best overall ranking on more than half of the datasets, and ranks second on the rest.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	66.59	85.31	98.57	33.15
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	<u>0.9293</u>	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

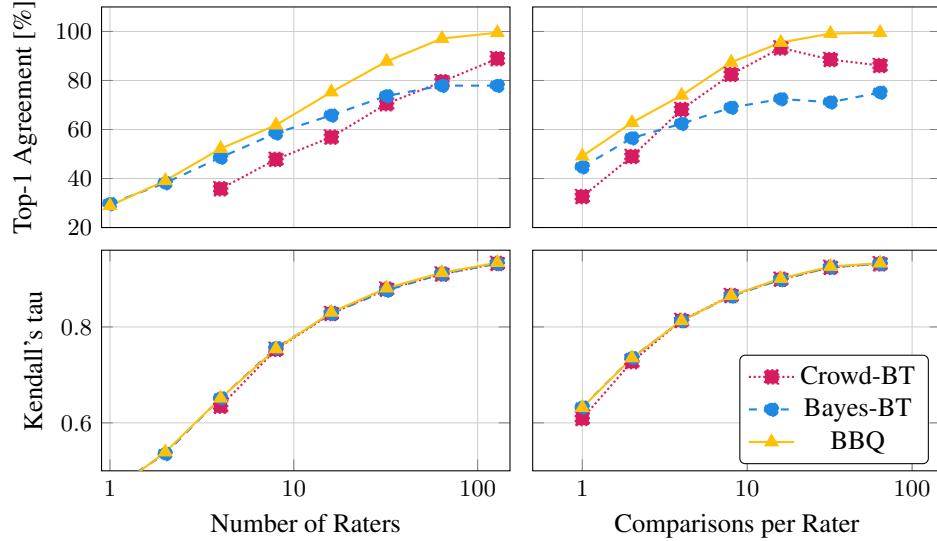


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** performance vs. number of raters. **Right:** performance vs. number of comparisons per rater. Both Top-1 agreement and Kendall’s τ improve steadily with more raters or comparisons. Top-1 agreement clearly separates the models, whereas Kendall’s τ stays similar across them. Crowd-BT fails to converge with very few raters, exposing a weakness that the EM algorithm avoids. Crowd-BT and BBQ track each other under sparse data, but BBQ pulls ahead of Bayes-BT as the number of raters or comparisons grows.

4.2 SCALING WITH RATERS AND COMPARISONS

As Table 1 shows, datasets with more comparisons per model yield better performance. The number of comparisons can be raised in two ways: by adding raters, or by having each rater perform more comparisons.

Figure 1 isolates the impact of both factors on the three BT variants. The left column plots performance against the number of raters (with comparisons per rater fixed at the maximum), and the right column plots performance against the number of comparisons per rater (with the number of raters fixed at the maximum). Top-1 accuracy separates the models clearly, while Kendall’s τ remains similar across them. Crowd-BT fails to converge with only one or two raters—a direct consequence of relying on gradient descent, and a failure the EM algorithm’s convergence guarantee rules out.

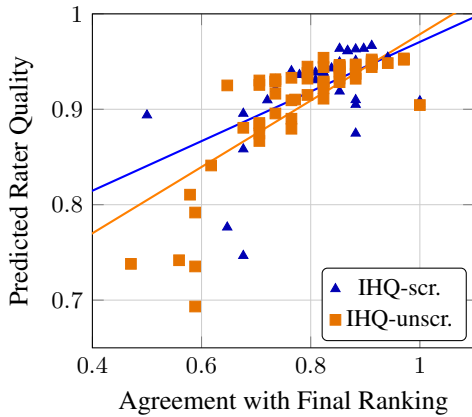


Figure 2: Rater agreement with the final ranking (x-axis) versus predicted rater quality (y-axis) for the IHQ datasets. Each point is an individual rater. Triangles denote the filtered dataset and squares the unfiltered dataset. Predicted quality tracks agreement closely, confirming that the model recovers rater reliability.

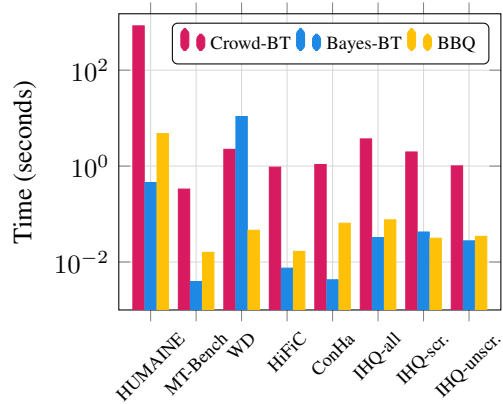


Figure 3: Average computation time in seconds (log scale) for the three models on a single bootstrapped sample across eight datasets. Crowd-BT can demand substantial computation on some datasets, whereas BBQ stays fast on every one.

Crowd-BT and BBQ perform comparably when the number of raters or comparisons per rater is small. BBQ needs both multiple raters and multiple comparisons per rater to distinguish rater qualities effectively, and once that data is available it increasingly outperforms Bayes-BT as either quantity grows. Bayes-BT underperforms overall, though it can edge out Crowd-BT when data per rater or the number of raters is small, since Crowd-BT cannot reliably estimate rater quality in such sparse settings.

4.3 RATER QUALITY

Figure 2 plots agreement with the final ranking against predicted rater quality. The relationship is a clear positive correlation: the more closely a rater agrees with the consensus ranking, the higher the quality the model assigns. For the filtered dataset the Pearson correlation is $r = 0.551$ across 50 raters, and for the unfiltered dataset it is stronger still, $r = 0.724$ across 62 raters.

The unscreened dataset, as expected, contains several raters with both lower agreement and lower predicted quality. Some IHQ-unscreened raters even fall below the level of random guessing (50% agreement), systematically disagreeing with the majority. This is exactly why modeling rater quality is essential when aggregating pairwise comparison data. BBQ pinpoints these raters, assigns them low quality scores, and thereby curtails their influence on the final ranking, neutralizing the noise they introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time in seconds each method needs to process a single bootstrapped sample across datasets. These timings speak directly to the feasibility of each method in real-world evaluation. BBQ converges within a few seconds on every dataset. Crowd-BT is slower than BBQ across the board and, on comparison-heavy datasets such as HUMAINE, becomes prohibitively slow—convergence takes roughly 15 minutes. Bayes-BT converges slowly on the WD dataset, though on other datasets it is marginally faster than BBQ. BBQ’s efficiency is a direct payoff of the closed-form EM updates we derive, which drive rapid convergence even on large datasets.

This comparison is in fact conservative: Crowd-BT is a highly optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ runs in plain NumPy with no specific optimization. BBQ therefore stands to become even faster with a compiled or vectorized implementation. BBQ is thus not only more robust and stable but also eminently practical for large-scale human preference

studies and any application requiring repeated bootstrapping. Its combination of accuracy, stability, and speed makes BBQ a compelling choice for real-world deployment.

4.5 CONFIDENCE INTERVALS

We next examine how Crowd-BT, Bayes-BT, and BBQ estimate uncertainty in item rankings. In a controlled simulation, raters provide pairwise comparisons of equally skilled items (H_0), and each model estimates skills and constructs uncertainty intervals to test for significant differences (Section G). The resulting Type I error rates appear in Figure 6. At the 99% confidence level the expected error rate is 1%. Crowd-BT and BBQ track this target closely, with slightly lower rates, while Bayes-BT is markedly conservative at $\sim 0.1\%$. BBQ therefore produces well-calibrated uncertainty estimates on par with Crowd-BT, whereas Bayes-BT substantially underestimates false positives.

5 LIMITATIONS

BBQ models rater quality effectively, but its advantage naturally narrows when all raters are uniformly reliable, since a homogeneous dataset leaves little variability to model. Guaranteeing consistently high-quality raters, though, is costly and often infeasible at scale. The practical implication is clear: BBQ is most valuable in realistic crowdsourced settings and matters less when evaluations are carefully curated.

A second boundary concerns the data the framework handles. BBQ currently targets pairwise comparisons, whereas many human evaluation studies use ratings, rankings, or multi-way inputs; extending the model to these forms of feedback is a natural next step that would broaden its applicability.

Methodologically, our experiments use a finite number of bootstrapped comparisons. This gives a practical measure of stability, and larger-scale studies would further validate robustness. The model also assumes independence across comparisons and does not model contextual or order effects, which can influence human judgments in practice.

Finally, although BBQ scales well, very large numbers of items or raters may still demand an optimized or compiled implementation.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Our core contribution is an Expectation-Maximization (EM) algorithm that estimates item skills and individual rater quality simultaneously, down-weighting or removing the influence of unreliable participants. By modeling rater quality explicitly, the method produces more stable and accurate rankings, and its EM formulation guarantees rapid convergence and monotonic likelihood improvement—overcoming the central limitation of gradient-based approaches.

Across diverse datasets, BBQ consistently achieved high Top-1 agreement and the strongest Kendall’s Tau, establishing both its robustness and its reliability. It excels precisely where evaluation is hardest—with noisy or crowdsourced raters—holding its accuracy even when raters contribute few comparisons. BBQ is especially effective at scale, where many raters are non-experts or vary widely in attentiveness and diligence, and our results make clear that incorporating rater quality is decisive whenever evaluation quality is heterogeneous or partially unreliable. The Bayesian framework further delivers principled uncertainty estimates for item scores, enabling interpretable comparisons across studies. We showed that the model’s error bars are well-calibrated and can be used to test whether differences between items are statistically significant, and that predicted rater quality aligns with agreement to the final rankings, confirming the model’s ability to identify reliable evaluators.

In sum, BBQ advances human evaluation methodology with a practical, interpretable, and generalizable approach to aggregating noisy pairwise comparisons, marking a significant step toward more cost-effective, interpretable, and reproducible human studies for evaluating AI-generated content.

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A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r has quality q_r : with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The derivation below is a standard EM (Dempster et al., 1977) argument that folds these rater-specific reliabilities directly into the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley–Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function w.r.t. the parameters (λ, q) , holding the expectations computed in the E-step fixed.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with the default hyperparameters.

The only hyperparameters in the Bayes-BT and BBQ models are the prior distribution parameters and the stopping thresholds. For both Bayes-BT and BBQ, we consider the model converged when no ELO score changes by more than 1 between two iterations. The ELO score can be calculated from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO.SCALE.FACTOR to 400.

We chose a gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the beta prior on the rater quality has $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate our models on a deliberately diverse set of human preference datasets spanning both language and image domains; Table 2 gives an overview.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides crowd-worker model comparison judgments on multi-turn dialogues.

For image compression, we use three datasets: WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which targets conditional generative models. These datasets span a wide range of scale, rater count, and rater expertise, giving a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected in a user study on the CLIC2024 (CLIC, 2025) data via the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To probe the impact of rater quality, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, which includes all raters. The screened subset captures higher-quality raters, while the unscreened subset faithfully reflects the noisy conditions typical of large-scale human evaluation.

Together, these datasets let us study both large-scale, relatively clean benchmarks and smaller, noisier settings where modeling rater reliability is decisive.

Table 2: Summary of the datasets used in our experiments. We collected the IHQ dataset on the Mabyduck platform in both 2AFC and 3AFC settings. For IHQ we provide screened and unscreened subsets: the screened subset includes only raters who passed pre-screening tests for attention and display quality, ensuring higher reliability, while the unscreened subset includes all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

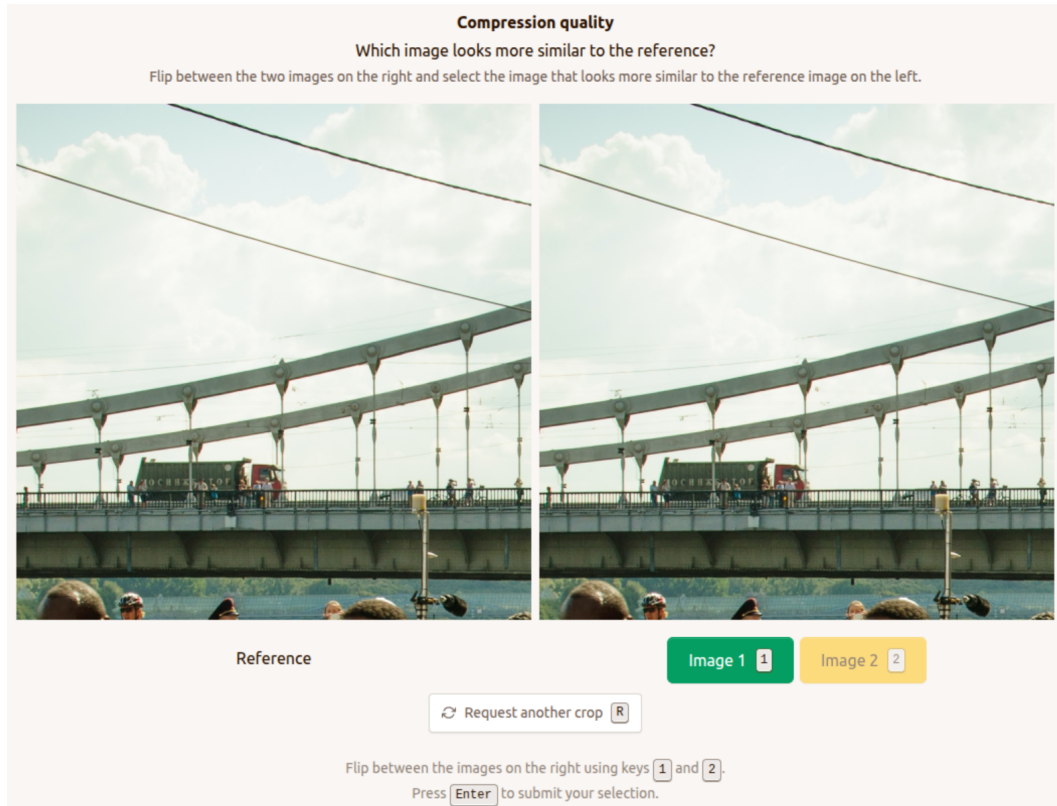


Figure 4: The Mabyduck user study platform used to collect pairwise comparisons. A reference image is shown on the left, and the rater selects between two compressed images on the right.

E USER STUDY PLATFORM

We collected all pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset using the Mabyduck platform (Ltd., 2025); Figure 4 shows a screenshot. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

To study rater quality, we split the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and completed training comparisons in which they chose the higher-quality image to reinforce the evaluation criteria.

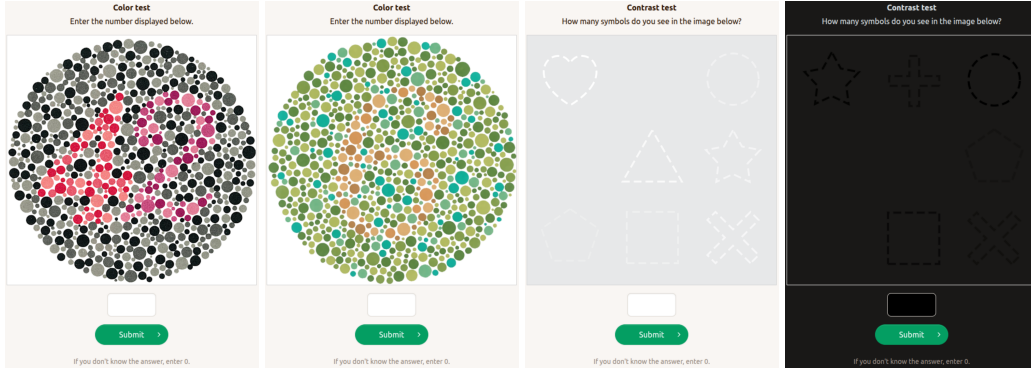


Figure 5: Four example images from the rater pre-screening process. The first two are color blindness tests, in which raters must identify the number displayed in each pattern. The last two are shape detection tests that measure sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background.

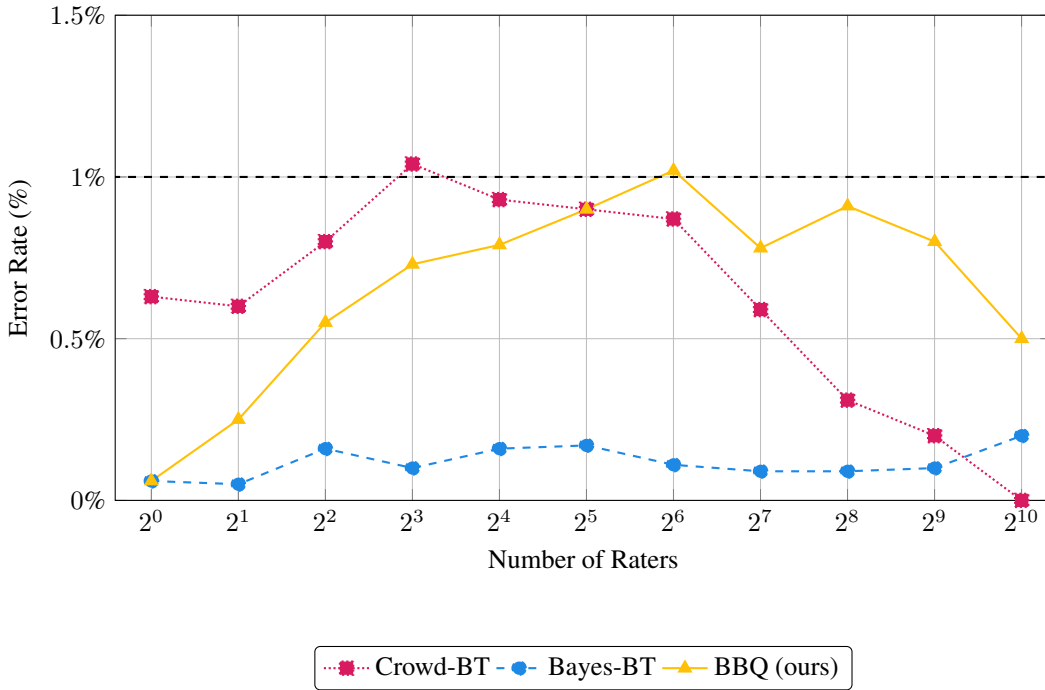


Figure 6: Type I error rate as a function of the number of raters. The dashed line marks the expected 1% error rate.

Figure 5 shows four example images from the pre-screening process. The first two are standard color blindness tests, in which raters must correctly identify the number shown in each pattern. The final two are shape detection tests that gauge sensitivity to low-contrast objects: one features a light gray shape on a white background, the other a dark gray shape on a black background. These tests ensure that only raters with adequate visual capabilities enter the screened subset.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT model rater quality explicitly, we bootstrap over raters rather than individual comparisons. This preserves each rater’s comparison distribution and yields a fair assessment of stability. We perform 10,000 bootstrap resamples of raters on every dataset except

HUMAINE, where a single Crowd-BT bootstrap iteration takes about 15 minutes (see Section 4.4); there we use 1,000 resamples to keep the computation tractable.

G UNCERTAINTY ESTIMATION DETAILS

Beyond ranking items, estimating uncertainty is essential for judging whether observed differences are statistically significant.

The Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) models quantify uncertainty through the posterior distribution over item skills. Each skill carries a Gamma prior that is updated with the observed pairwise comparisons, and credible intervals derived from the posterior are converted to the Elo scale to facilitate comparison across items.

The classical Crowd-BT model, being non-Bayesian, estimates uncertainty with a frequentist approximation: a second-order Taylor expansion around the maximum likelihood estimate. Inverting the Hessian of the log-likelihood yields the covariance matrix, whose diagonal entries give the per-item variances. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to the 99% confidence level. Narrower likelihood peaks yield smaller intervals, reflecting higher certainty, while flatter peaks produce larger intervals, reflecting greater uncertainty.

To measure how well these models estimate uncertainty, we run a controlled simulation experiment. We generate trials with two items of equal skill (50–50 win probability) and simulate pairwise comparison data by coin flips. For each trial we simulate multiple raters (users) and apply each model to estimate Elo scores and their uncertainty.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial we compute the 99% confidence (or credible) interval for each item’s skill estimate. A model incorrectly rejects H_0 when the confidence intervals do not overlap, signaling a statistically significant difference where none exists. Our primary metric is the frequency with which each model wrongly concludes that the items differ—the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater, varying the number of raters to see how uncertainty estimates scale with data. This lets us estimate the empirical Type I error rate for each model and compare it against the theoretical 1% at the 99% confidence level. As Figure 6 shows, Crowd-BT and BBQ hold Type I error rates close to the expected 1%, confirming that their uncertainty estimates are well-calibrated. Bayes-BT, in contrast, is overly conservative, consistently producing lower error rates than expected—a sign that its credible intervals are poorly calibrated for significance testing.